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Java SE 27 & JDK 27

Class SelectorProvider

java.lang.Object
    java.nio.channels.spi.SelectorProvider

```
public abstract class SelectorProvider
extends Object
```

Service-provider class for selectors and selectable channels.

A selector provider is a concrete subclass of this class that has a zero-argument constructor and implements the abstract methods specified below. A given invocation of the Java virtual machine maintains a single system-wide default provider instance, which is returned by the `provider` method. The first invocation of that method will locate the default provider as specified below.

The system-wide default provider is used by the static open methods of the `DatagramChannel`, `Pipe`, `Selector`, `ServerSocketChannel`, and `SocketChannel` classes. It is also used by the `System.inheritedChannel()` method. A program may make use of a provider other than the default provider by instantiating that provider and then directly invoking the open methods defined in this class.

All
of the methods in this class are safe for use by multiple concurrent threads.

Since:

1.4

Constructor Summary

Constructors

Modifier	Constructor	Description
protected	`SelectorProvider()`	Initializes a new instance of this class.

Method Summary

All Methods | **Static Methods**

Modifier and Type	Method	Description
`Channel`	`inheritedChannel()`	Returns the channel inherited from the entity that created this Java virtual machine.

abstract DatagramChannel	**openDatagramChannel**()	Opens a datagram channel.
abstract DatagramChannel	**openDatagramChannel** (ProtocolFamily family)	Opens a datagram channel.
abstract Pipe	**openPipe**()	Opens a pipe.
abstract AbstractSelector	**openSelector**()	Opens a selector.
abstract ServerSocketChannel	**openServerSocketChannel**(	Opens a server-socket channel.
ServerSocketChannel	**openServerSocketChannel** (ProtocolFamily family)	Opens a server-socket channel.
abstract SocketChannel	**openSocketChannel**()	Opens a socket channel.
SocketChannel	**openSocketChannel** (ProtocolFamily family)	Opens a socket channel.
static SelectorProvider	**provider**()	Returns the system-wide default selector provider for this invocation of the Java virtual machine.

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructor Details

SelectorProvider

```
protected SelectorProvider()
```

Initializes a new instance of this class.

Method Details

provider

```
public static SelectorProvider provider()
```

Returns the system-wide default selector provider for this invocation of the Java virtual machine.

The first invocation of this method locates the default provider object as follows:

1. If the system property `java.nio.channels.spi.SelectorProvider` is defined then it is taken to be the fully-qualified name of a concrete provider class. The class is loaded and instantiated; if this process fails then an unspecified error is thrown.

2. If a provider class has been installed in a jar file that is visible to the system class loader, and that jar file contains a provider-configuration file named `java.nio.channels.spi.SelectorProvider` in the resource directory `META-INF/services`, then the first class name specified in that file is taken. The class is loaded and instantiated; if this process fails then an unspecified error is thrown.

3. Finally, if no provider has been specified by any of the above means then the system-default provider class is instantiated and the result is returned.

Subsequent invocations of this method return the provider that was returned by the first invocation.

Returns:

The system-wide default selector provider

openDatagramChannel

```
public abstract DatagramChannel openDatagramChannel()
                                                throws IOException
```

Opens a datagram channel.

Returns:

The new channel

Throws:

`IOException` - If an I/O error occurs

openDatagramChannel

```
public abstract DatagramChannel openDatagramChannel
(ProtocolFamily family)
                                                throws IOException
```

Opens a datagram channel.

Parameters:

`family` - The protocol family

Returns:

A new datagram channel

Throws:

`UnsupportedOperationException` - If the specified protocol family is not supported

`IOException` - If an I/O error occurs

Since:

1.7

openPipe

```
public abstract Pipe openPipe()
                        throws IOException
```

Opens a pipe.

Returns:

The new pipe

Throws:

`IOException` - If an I/O error occurs

openSelector

```
public abstract AbstractSelector openSelector()
                                          throws IOException
```

Opens a selector.

Returns:

The new selector

Throws:

`IOException` - If an I/O error occurs

openServerSocketChannel

```
public abstract ServerSocketChannel openServerSocketChannel()
                                                           throws
IOException
```

Opens a server-socket channel.

Returns:

The new channel

Throws:

IOException - If an I/O error occurs

openSocketChannel

```
public abstract SocketChannel openSocketChannel()
                                            throws IOException
```

Opens a socket channel.

Returns:

The new channel

Throws:

IOException - If an I/O error occurs

inheritedChannel

```
public Channel inheritedChannel()
                          throws IOException
```

Returns the channel inherited from the entity that created this Java virtual machine.

On many operating systems a process, such as a Java virtual machine, can be started in a manner that allows the process to inherit a channel from the entity that created the process. The manner in which this is done is system dependent, as are the possible entities to which the channel may be connected. For example, on UNIX systems, the Internet services daemon (*inetd*) is used to start programs to service requests when a request arrives on an associated network port. In this example, the process that is started, inherits a channel representing a network socket.

In cases where the inherited channel is for an *Internet protocol* socket then the Channel type returned by this method is determined as follows:

- If
  the inherited channel is for a stream-oriented connected socket then a SocketChannel is returned. The socket channel is, at least initially, in blocking mode, bound to a socket address, and connected to a peer.
- If the inherited channel is for a stream-oriented listening socket then a ServerSocketChannel is returned. The server-socket channel is, at least initially, in blocking mode, and bound to a socket address.
- If the inherited channel is a datagram-oriented socket then a DatagramChannel is returned. The datagram channel is, at least initially, in blocking mode, and bound to a socket address.

In cases where the inherited channel is for a *Unix domain* socket then the `Channel` type returned is the same as for *Internet protocol* sockets as described above, except that datagram-oriented sockets are not supported.

In addition to the two types of socket just described, this method may return other types in the future.

The first invocation of this method creates the channel that is returned. Subsequent invocations of this method return the same channel.

Implementation Requirements:

The default implementation of this method returns `null`.

Returns:

The inherited channel, if any, otherwise `null`.

Throws:

`IOException` - If an I/O error occurs

Since:

1.5

openSocketChannel

```
public SocketChannel openSocketChannel(ProtocolFamily family)
                                throws IOException
```

Opens a socket channel.

Implementation Requirements:

The default implementation of this method first checks that the given protocol `family` is not `null`, then throws `UnsupportedOperationException`.

Parameters:

`family` - The protocol family

Returns:

The new channel

Throws:

`UnsupportedOperationException` - If the specified protocol family is not supported

`IOException` - If an I/O error occurs

Since:

15

openServerSocketChannel

```
public ServerSocketChannel openServerSocketChannel
(ProtocolFamily family)
                                                throws IOException
```

Opens a server-socket channel.

Implementation Requirements:

The default implementation of this method first checks that the given protocol `family` is not `null`, then throws `UnsupportedOperationException`.

Parameters:

`family` - The protocol family

Returns:

The new channel

Throws:

`UnsupportedOperationException` - If the specified protocol family is not supported

`IOException` - If an I/O error occurs

Since:

15

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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